

Summary valuation methods

QF*

Starting point is a model for the ‘basic assets’ in the market, i.e. SDEs. In this summary we assume that these assets are a stock, with price process S , and a money-market account, with price process B . Given this model, this brief note summarizes the three methods to value a European option with payoff $C_T = h(S_T)$ at maturity T .

1 Partial differential equations

1. Restrict yourself to Markov trading strategies. So we can write

$$V_t = F(t, S_t). \quad (1)$$

2. Restrict yourself to self-financing trading strategies. So we have

$$dV_t = \phi_t dS_t + \psi_t dB_t. \quad (2)$$

3. Derive representation for dynamics of V_t :

- using (1) (and Itô’s formula) we obtain $dV_t = a_t dt + b_t dW_t$;
- using (2) we obtain $dV_t = c_t dt + d_t dW_t$.

4. Solving $a_t = c_t$ and $b_t = d_t$ yields a PDE for F .

5. To value the European option with payoff $C_T = h(S_T)$: solve the PDE from step 4 under the terminal condition $F(T, s) = h(s)$ for all $s > 0$. The fair price, at time t , is then given by $C_t = F(t, S_t)$.

2 Risk-neutral world

1. Pick a numéraire N (any of the basic assets with $N_t > 0$).
2. By the risk-neutral version of the First Fundamental Theorem of asset pricing (FFT) our basic market is free of arbitrage opportunities “if and only if” there exists a probability measure $\mathbb{Q}$ such that $\mathbb{Q} \sim \mathbb{P}$ and such that for all assets the discounted price, i.e. A_t/N_t , is a martingale under $\mathbb{Q}$.
3. Use the model to derive, for every basic asset, a SDE for the discounted price process, i.e.

$$d\frac{A_t}{N_t} = a_t dt + b_t dW_t^{\mathbb{P}}.$$

*If you have comments/questions please send an e-mail to rakker@uvt.nl.

4. Introduce a new process $W^{\mathbb{Q}}$ by (we specify γ_t at a later stage)

$$dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}}, \quad W_0^{\mathbb{Q}} = 0,$$

and insert this into the previous display:

$$d\frac{A_t}{N_t} = (a_t - b_t\gamma_t) dt + b_t dW_t^{\mathbb{Q}}.$$

Choose $\gamma_t = a_t/b_t$, which yields:

$$d\frac{A_t}{N_t} = b_t dW_t^{\mathbb{Q}}.$$

(For the Black-Scholes market with $N = B$ this gives $\gamma_t = (\mu - r)/\sigma$, the market price of risk.)

5. Apply Girsanov's theorem to find a probability measure $\mathbb{Q}$ that is equivalent to $\mathbb{P}$ and such that $W^{\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$.
6. Conclude that all A_t/N_t are martingales under $\mathbb{Q}$.
7. To price the European option with payoff $C_T = h(S_T)$:
 - solving the SDEs (under $\mathbb{Q}$) typically yields $S_T = g(T, W_T^{\mathbb{Q}})$;
 - apply the FFT once more:

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[\frac{1}{N_T} h(g(T, W_T^{\mathbb{Q}})) \mid \mathcal{F}_t \right],$$

$$\text{in particular } C_0 = N_0 \mathbb{E}_{\mathbb{Q}} \left[\frac{1}{N_T} h(g(T, W_T^{\mathbb{Q}})) \right],$$

and exploit that $W^{\mathbb{Q}}$ is a standard Brownian motion under $\mathbb{Q}$.

3 Pricing kernel

By the pricing kernel version of the FFT our market is free of arbitrage opportunities “if and only if” there exists some adapted process K_t such that $K_t > 0$ and such that for all basic assets in the market $K_t A_t$ is a martingale (here A_t denotes the price of the asset).

1. Try $dK_t = \mu_t^K dt + \sigma_t^K dW_t^{\mathbb{P}}$.
2. Use the model for the basic assets to solve for μ_t^K and σ_t^K , i.e. calculate $dK_t A_t = a_t dt + b_t dW_t^{\mathbb{P}}$ and ensure $a_t = 0$ for every basic asset.
3. To price the European option with payoff $C_T = h(S_T)$ apply the FFT once more:

$$C_t = \frac{1}{K_t} \mathbb{E}_{\mathbb{P}}[K_T C_T \mid \mathcal{F}_t], \quad \text{in particular } C_0 = \frac{1}{K_0} \mathbb{E}_{\mathbb{P}}[K_T C_T].$$